The Double Factorial and Its Inverse

Complete Asymptotic Transseries, Borel Summation, and General Coefficient Formulae

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Abstract

The double factorial sequence $n!!$ is not the restriction of one canonical slowly varying function: its even and odd subsequences are two gamma-function branches whose quotient contains a non-vanishing parity factor. This article therefore treats three related inverse problems separately and exactly: the two parity-resolved smooth inverses, the integer-valued generalized inverse, and the eventual inverse of the periodic analytic continuation recorded in OEIS A006882.

For $r \in \{0,1\}$ we introduce

$$\mathcal{D}_r(x) = \kappa_r 2^{x/2}\Gamma\Big(\frac{x}{2}+1\Big), \qquad \kappa_0 = 1, \qquad \kappa_1 = \sqrt{\frac{2}{\pi}},$$

so that $\mathcal{D}_r(n) = n!!$ whenever $n \equiv r \pmod 2$. With $s = x+1$ and $C_0 = \frac{1}{2}\log\pi$, $C_1 = \frac{1}{2}\log 2$, the optimally centred logarithm is

$$\log\mathcal{D}_r(s-1) \sim \frac{s}{2}(\log s - 1) + C_r + \sum_{k\ge 1}\frac{\beta_k}{s^{2k-1}}, \qquad \beta_k = \frac{(1-2^{2k-1})B_{2k}}{2k(2k-1)}.$$

We give all-order additive and multiplicative coefficient formulae in terms of Bernoulli numbers, integer partitions, and complete Bell polynomials. More strongly, the divergent logarithmic series is the asymptotic expansion of the exact Borel–Laplace integral

$$\int_0^\infty \mathrm{e}^{-st}\left(\frac{1}{2t\sinh t} - \frac{1}{2t^2}\right)\mathrm{d}t.$$

This supplies alternating remainder bounds, the complete Borel singularity lattice, factorial large-order estimates, and the optimal-truncation scale.

For the large real inverse, put

$$R_r = \log y - C_r, \qquad w_r = \mathrm{W}_0(2R_r/\mathrm{e}), \qquad T_r = \frac{2R_r}{w_r}, \qquad \Lambda_r = w_r + 1 = \log T_r.$$

Then, for every fixed N,

$$\mathcal{D}_r^{-1}(y) = T_r - 1 + \sum_{n=1}^{N}\frac{d_{2n-1}(\Lambda_r)}{T_r^{2n-1}} + O\left(\frac{1}{T_r^{2N+1}\Lambda_r}\right).$$

The coefficient d_{2n-1} is a rational polynomial in Λ^{-1} of degree at most $2n-1$. We derive both a triangular recurrence and a finite, non-recursive Lagrange–Bürmann formula. The latter is expressed through partial Bell polynomials and proves the global sign law $(-1)^{n+1}d_{2n-1} \in \Lambda^{-1}\mathbb{Q}_{>0}[\Lambda^{-1}]$. Explicit blocks are listed through T^{-9}, and the Lambert core is expanded to all orders in iterated logarithms using unsigned Stirling numbers of the first kind.

Finally, for the OEIS continuation

$$\mathfrak{D}(x) = 2^{x/2}\Gamma\Big(\frac{x}{2}+1\Big)\left(\frac{2}{\pi}\right)^{(1-\cos\pi x)/4},$$

the inverse has a leading log-periodic sector of order $1/\log T$, larger than all inverse-power corrections. We give its closed Fourier–Lagrange coefficients and a second triangular recurrence that couples this periodic sector to the Bernoulli grid. Thus the ambiguity in “the inverse double factorial” is resolved rather than hidden.

Keywords. Double factorial; inverse double factorial; Lambert W; asymptotic transseries; Stirling series; Borel summation; Bernoulli numbers; Bell polynomials; Lagrange–Bürmann inversion; parity; log-periodic asymptotics.

Contents

1 Introduction

The double factorial is defined on nonnegative integers by

$$0!! = 1!! = 1, \qquad n!! = n(n-2)!! \quad (n \geq 2). \tag{1}$$

Equivalently, $n!!$ is the product of the positive integers not exceeding n that have the same parity as n. The sequence begins

$$1, 1, 2, 3, 8, 15, 48, 105, 384, 945, \ldots$$

and is OEIS A006882 [1]. Its elementary closed forms are

$$(2m)!! = 2^m m!, \qquad (2m-1)!! = \frac{(2m)!}{2^m m!} = \frac{2^m \Gamma(m+1/2)}{\sqrt{\pi}}. \tag{2}$$

These identities already expose the main asymptotic issue: the even and odd subsequences have different multiplicative constants of leading order. A single ordinary Poincaré expansion with constant coefficients cannot represent both without retaining a parity monomial such as $(-1)^n$.

The word *inverse* introduces a second issue. A sequence has no ordinary inverse function until an interpolation or a generalized-inverse convention is chosen. There are three natural choices here.

1. The even and odd subsequences have canonical gamma continuations $\mathcal{D}_0$ and $\mathcal{D}_1$. Their smooth inverses are the cleanest objects for high-accuracy asymptotics and preserve the exact parity normalization.
2. The integer sequence has a right-continuous generalized inverse, obtained by applying a parity-preserving floor to the two smooth inverses.
3. OEIS records a single analytic continuation $\mathfrak{D}$ which agrees with both parities. It contains a periodic factor, and its eventual inverse has a log-periodic transseries qualitatively different from either $\mathcal{D}_r^{-1}$.

The article develops all three and makes their relationship explicit.

Two asymptotic scales must also be separated. Taking the logarithm converts the main growth into

$$\frac{s}{2}(\log s - 1), \qquad s = x + 1.$$

Its inverse is exactly a Lambert-W expression. Expanding W produces an *outer* series in powers of $1/\log\log y$ with polynomial dependence on $\log\log\log y$. The Bernoulli corrections generate a much finer grid in odd inverse powers of the Lambert core T. The fixed shift -1 lies between these levels: it is beyond every finite order of the outer Lambert expansion, but much larger than $1/(T \log T)$. Treating all these terms as one ordinary series loses the natural well-ordering of the transseries.

Interpretive point

There is no contradiction between the two statements

$$n!! \sim A_n \sqrt{n}\,(n/\mathrm{e})^{n/2} \quad \text{and} \quad n!! \sim A_n \big((n+1)/\mathrm{e}\big)^{(n+1)/2}.$$

They use different normal forms. The first leaves a factor $\sqrt{n}$ and has an uncentred Stirling series. The second absorbs that factor by the optimal shift $s = n + 1$ and has an odd inverse-power logarithmic series. The centred form is far better for inversion.

1.1 Principal formulae

Put

$$\kappa_0 = 1, \qquad \kappa_1 = \sqrt{2/\pi}, \qquad C_0 = \frac{1}{2}\log\pi, \qquad C_1 = \frac{1}{2}\log 2. \tag{3}$$

For $r \in \{0,1\}$ define

$$\mathcal{D}_r(x) = \kappa_r 2^{x/2}\Gamma\left(\frac{x}{2}+1\right). \tag{4}$$

The direct logarithmic coefficients in the centred variable $s = x+1$ are

$$\beta_k = \frac{(1-2^{2k-1})B_{2k}}{2k(2k-1)} = (-1)^k \frac{(2k-2)!\,\eta(2k)}{\pi^{2k}}, \tag{5}$$

where $\eta(z) = (1-2^{1-z})\zeta(z)$ is the Dirichlet eta function. Let

$$\alpha_k = 2\beta_k = \frac{(1-2^{2k-1})B_{2k}}{k(2k-1)}. \tag{6}$$

Main result

For $y \to +\infty$ and $r \in \{0,1\}$, define

$$R_r = \log y - C_r, \qquad w_r = \mathrm{W}_0(2R_r/\mathrm{e}), \qquad T_r = \frac{2R_r}{w_r} = \mathrm{e}^{w_r+1}, \qquad \Lambda_r = w_r + 1. \tag{7}$$

Then

$$\mathcal{D}_r^{-1}(y) \sim T_r - 1 + \sum_{n\geq 1} \frac{d_{2n-1}(\Lambda_r)}{T_r^{2n-1}}, \tag{8}$$

where $d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$ and

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \tag{9}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2+10\Lambda+5}{360\Lambda^3}, \tag{10}$$

$$d_5(\Lambda) = \frac{2232\Lambda^4+1176\Lambda^3+714\Lambda^2+350\Lambda+105}{45360\Lambda^5}. \tag{11}$$

Every coefficient is generated by the recurrence in Theorem 7.1; the finite closed formula is Theorem 8.1. Expanding the retained Lambert function gives an outer iterated-logarithmic transseries with coefficients generated from unsigned Stirling numbers; see Section 9.

2 Exact continuations and the meaning of the inverse

2.1 The two parity-resolved gamma branches

Proposition 2.1 (Exact parity interpolation). *For every integer $n \geq 0$ and $r \in \{0,1\}$ with $n \equiv r \pmod 2$,*

$$\mathcal{D}_r(n) = n!!. \tag{12}$$

Moreover, $\mathcal{D}_r$ is strictly increasing on $[0,\infty)$, and hence has a single-valued inverse on its range.

Proof. If $n = 2m$, then $\mathcal{D}_0(2m) = 2^m\Gamma(m+1) = 2^m m! = (2m)!!$. If $n = 2m+1$, then

$$\mathcal{D}_1(2m+1) = \sqrt{\frac{2}{\pi}}\,2^{m+1/2}\Gamma(m+3/2) = \frac{2^{m+1}}{\sqrt{\pi}}\Gamma(m+3/2) = (2m+1)!!.$$

For real $x \geq 0$,

$$\frac{\mathcal{D}_r'(x)}{\mathcal{D}_r(x)} = \frac{1}{2}\left(\log 2 + \psi\left(\frac{x}{2}+1\right)\right), \tag{13}$$

where $\psi = \Gamma'/\Gamma$. The trigamma series $\psi'(z) = \sum_{m\geq 0}(z+m)^{-2}$ is positive for $z > 0$, so the right-hand side is increasing. At $x = 0$ it equals $\frac{1}{2}(\log 2 - \gamma) > 0$. Thus it remains positive. □

Remark 2.2. The factor κ_r affects only the target normalization; it disappears from the logarithmic derivative. Consequently both parity branches have the same local conditioning and the same inverse correction polynomials. Their only asymptotic difference is the constant C_r absorbed into R_r.

2.2 The single periodic OEIS continuation

A convenient one-function interpolation is

$$\mathfrak{D}(x) = 2^{x/2}\Gamma\Big(\frac{x}{2}+1\Big)\left(\frac{2}{\pi}\right)^{(1-\cos\pi x)/4}. \tag{14}$$

For even integers the periodic factor is 1, while for odd integers it is $\sqrt{2/\pi}$; hence $\mathfrak{D}(n) = n!!$ for all $n \geq 0$. This is the analytic extension recorded in the OEIS entry [1].

Let

$$C_* = \frac{1}{4}\log(2\pi), \qquad a = \frac{1}{4}\log\frac{\pi}{2} > 0. \tag{15}$$

In the centred variable $s = x+1$, the exact relation developed below is

$$\log\mathfrak{D}(s-1) = \frac{s}{2}(\log s - 1) + C_* - a\cos(\pi s) + \mathcal{C}(s), \tag{16}$$

where $\mathcal{C}$ is the Borel-summed centred Stirling correction. Thus the periodic term remains $O(1)$ in the logarithm. On inversion it produces an $O(1/\log T)$ displacement, which is much larger than the $O(1/(T\log T))$ first Bernoulli displacement.

The function $\mathfrak{D}$ is eventually strictly increasing. Indeed,

$$\frac{\mathfrak{D}'(x)}{\mathfrak{D}(x)} = \frac{1}{2}\left(\log 2 + \psi\Big(\frac{x}{2}+1\Big)\right) + \frac{\pi}{4}\log\frac{2}{\pi}\,\sin(\pi x). \tag{17}$$

The first term grows like $\frac{1}{2}\log x$, while the second is bounded. More explicitly, the standard bound $\psi(z) > \log(z-1/2)$ for $z > 1/2$ gives a lower bound $\frac{1}{2}\log(x+1) - \pi a$, which is positive for all sufficiently large x. The eventual inverse $\mathfrak{D}_+^{-1}$ is therefore well defined; its complete formal construction is given in Section 10.

2.3 The discrete generalized inverse

For a fixed parity, define the largest same-parity index whose double factorial does not exceed y by

$$J_r(y) = r + 2\left\lfloor \frac{\mathcal{D}_r^{-1}(y) - r}{2}\right\rfloor, \qquad r \in \{0,1\}, \tag{18}$$

whenever y is in the relevant range. The generalized inverse of the whole sequence is then

$$J(y) = \max\{J_0(y), J_1(y)\}. \tag{19}$$

The smooth transseries computed later can be inserted into Equation (18). Because the final floor is discontinuous, a rigorous integer answer requires an error bound smaller than the distance to the nearest parity threshold; this is discussed in Section 13.

3 Direct asymptotics: two optimal normal forms

Throughout, Bernoulli numbers are normalized by

$$\frac{t}{\mathrm{e}^t - 1} = \sum_{n\geq 0} B_n \frac{t^n}{n!}, \qquad B_1 = -\frac{1}{2}. \tag{20}$$

The classical Stirling series is

$$\log \Gamma(z) \sim \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2}\log(2\pi) + \sum_{k\geq 1} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}, \tag{21}$$

valid in closed subsectors of $|\arg z| < \pi$ [2].

3.1 Uncentred expansion

Theorem 3.1 (Uncentred logarithmic expansion). *As $x \to +\infty$,*

$$\log \mathcal{D}_r(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2}\log x + C_r + \sum_{k\geq 1} \frac{\delta_k}{x^{2k-1}}, \tag{22}$$

where

$$\delta_k = \frac{2^{2k-1}B_{2k}}{2k(2k-1)} = (-1)^{k-1}\frac{(2k-2)!\zeta(2k)}{\pi^{2k}}. \tag{23}$$

In particular,

$$\delta_1 = \frac{1}{6}, \quad \delta_2 = -\frac{1}{45}, \quad \delta_3 = \frac{8}{315}, \quad \delta_4 = -\frac{8}{105}, \quad \delta_5 = \frac{128}{297}. \tag{24}$$

Proof. Apply Equation (21) to $\Gamma(x/2+1)$ in the equivalent shifted form

$$\log \Gamma(z+1) \sim \left(z + \frac{1}{2}\right) \log z - z + \frac{1}{2}\log(2\pi) + \sum_{k\geq 1} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}$$

with $z = x/2$, and add $(x/2)\log 2 + \log \kappa_r$. The elementary terms reduce to the first three terms in Equation (22), while $z^{-(2k-1)} = 2^{2k-1}x^{-(2k-1)}$. Euler's evaluation of $\zeta(2k)$ gives the second form of δ_k. □

Exponentiating gives

$$\mathcal{D}_r(x) \sim A_r\sqrt{x}\left(\frac{x}{\mathrm{e}}\right)^{x/2} \sum_{n\geq 0} \frac{g_n}{x^n}, \qquad A_0 = \sqrt{\pi}, \quad A_1 = \sqrt{2}. \tag{25}$$

The coefficients g_n are developed in Section 4.

3.2 The optimally centred expansion

The shift $s = x + 1$ removes the explicit square-root factor and annihilates all even inverse powers in the logarithm.

Theorem 3.2 (Centred logarithmic expansion). *Let $s = x + 1$. Then*

$$\log \mathcal{D}_r(s-1) \sim \frac{s}{2}(\log s - 1) + C_r + \sum_{k\geq 1} \frac{\beta_k}{s^{2k-1}}, \tag{26}$$

where β_k is given by Equation (5). The first terms are

$$\beta_1 = -\frac{1}{12}, \quad \beta_2 = \frac{7}{360}, \quad \beta_3 = -\frac{31}{1260}, \quad \beta_4 = \frac{127}{1680}, \quad \beta_5 = -\frac{511}{1188}. \tag{27}$$

Equivalently,

$$\mathcal{D}_r(s-1) \sim A_r\left(\frac{s}{\mathrm{e}}\right)^{s/2} \sum_{n\geq 0} \frac{c_n}{s^n}. \tag{28}$$

Proof. The duplication formula

$$\Gamma(s) = \frac{2^{s-1}}{\sqrt{\pi}} \Gamma\left(\frac{s}{2}\right) \Gamma\left(\frac{s+1}{2}\right) \tag{29}$$

(see, for example, [3]) rewrites Equation (4) as

$$\mathcal{D}_r(s-1) = \kappa_r \sqrt{\pi}\, 2^{(1-s)/2} \frac{\Gamma(s)}{\Gamma(s/2)}. \tag{30}$$

Subtracting the two Stirling expansions cancels the logarithmic square-root term. The kth correction is

$$\frac{B_{2k}}{2k(2k-1)} \left(s^{-(2k-1)} - (s/2)^{-(2k-1)}\right),$$

which is precisely $\beta_k s^{-(2k-1)}$. The constant is $\log(\kappa_r \sqrt{\pi}) = C_r$. Exponentiation gives Equation (28). □

At integer n, the two parity constants can be written in one expression:

$$A_n = (2\pi)^{1/4} \left(\frac{\pi}{2}\right)^{(-1)^n/4} = \pi^{(1+(-1)^n)/4} 2^{(1-(-1)^n)/4}. \tag{31}$$

Hence the unified centred sequence expansion is

$$n!! \sim A_n \left(\frac{n+1}{\mathrm{e}}\right)^{(n+1)/2} \sum_{j \geq 0} \frac{c_j}{(n+1)^j}. \tag{32}$$

The periodic amplitude is not an error term; it is part of the leading transmonomial.

4 Combinatorial formulae for the direct coefficients

The passage from the logarithmic series to the multiplicative series is an instance of the exponential formula. We record it in three equivalent forms: a partition sum, a complete Bell polynomial, and a triangular recurrence. This makes every direct coefficient computable without re-expanding an exponential from scratch.

4.1 Bell-polynomial conventions

The exponential partial Bell polynomials are defined by

$$\frac{1}{k!} \left(\sum_{m \geq 1} x_m \frac{z^m}{m!}\right)^k = \sum_{n \geq k} \mathrm{B}_{n,k}(x_1, \ldots, x_{n-k+1}) \frac{z^n}{n!}, \tag{33}$$

and the complete Bell polynomial is

$$\mathrm{Y}_n(x_1, \ldots, x_n) = \sum_{k=0}^{n} \mathrm{B}_{n,k}(x_1, \ldots, x_{n-k+1}). \tag{34}$$

Thus

$$\exp\left(\sum_{m \geq 1} x_m \frac{z^m}{m!}\right) = \sum_{n \geq 0} \mathrm{Y}_n(x_1, \ldots, x_n) \frac{z^n}{n!}. \tag{35}$$

These conventions agree with the coefficient calculus used in [11, 7].

Theorem 4.1 (All direct multiplicative coefficients). *Define sparse logarithmic coefficient sequences*

$$\ell_{2k-1}^{(u)} = \delta_k, \quad \ell_{2k}^{(u)} = 0, \qquad \ell_{2k-1}^{(c)} = \beta_k, \quad \ell_{2k}^{(c)} = 0. \tag{36}$$

Then the coefficients in Equation (25) and Equation (28) are

$$g_n = \frac{1}{n!}\mathrm{Y}_n\bigl(1!\ell_1^{(u)}, 2!\ell_2^{(u)}, \dots, n!\ell_n^{(u)}\bigr), \tag{37}$$

$$c_n = \frac{1}{n!}\mathrm{Y}_n\bigl(1!\ell_1^{(c)}, 2!\ell_2^{(c)}, \dots, n!\ell_n^{(c)}\bigr). \tag{38}$$

Equivalently,

$$g_n = \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_{k\geq 1}(2k-1)m_k = n}} \prod_{k\geq 1} \frac{\delta_k^{m_k}}{m_k!}, \tag{39}$$

$$c_n = \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_{k\geq 1}(2k-1)m_k = n}} \prod_{k\geq 1} \frac{\beta_k^{m_k}}{m_k!}. \tag{40}$$

They obey the recurrences

$$g_0 = 1, \qquad n g_n = \sum_{1\leq k\leq (n+1)/2} (2k-1)\delta_k \, g_{n-2k+1}, \tag{41}$$

$$c_0 = 1, \qquad n c_n = \sum_{1\leq k\leq (n+1)/2} (2k-1)\beta_k \, c_{n-2k+1}. \tag{42}$$

Proof. Put $z = x^{-1}$ in the uncentred expansion. The correction factor is

$$\exp\left(\sum_{k\geq 1} \delta_k z^{2k-1}\right).$$

The formula in Equation (37) follows from Equation (35), and expanding the exponential as a product of $\sum_{m_k \geq 0} \delta_k^{m_k} z^{(2k-1)m_k}/m_k!$ gives Equation (39). If $G(z) = \sum_{n\geq 0} g_n z^n$, logarithmic differentiation gives

$$zG'(z) = G(z)\sum_{k\geq 1}(2k-1)\delta_k z^{2k-1}.$$

Coefficient extraction proves Equation (41). Replacing δ_k by β_k proves all centred statements. ☐

The first coefficients are as follows.

Table 1: Multiplicative coefficients in the uncentred and centred normal forms.

n	g_n	c_n
0	1	1
1	$1/6$	$-1/12$
2	$1/72$	$1/288$
3	$-139/6480$	$1003/51840$
4	$-571/155520$	$-4027/2488320$
5	$163879/6531840$	$-5128423/209018880$
6	$5246819/1175731200$	$168359651/75246796800$
7	$-534703531/7054387200$	$68168266699/902961561600$
8	$-4483131259/338610585600$	$-587283555451/86684309913600$

Remark 4.2 (Arithmetic). All g_n and c_n are rational because the logarithmic coefficients are rational. The Bell formula is preferable for structural arguments, while the recurrence is normally preferable for exact computation: it produces the first N coefficients with $O(N^2)$ rational operations and exposes the sparse odd support directly.

5 Exact Borel sums, remainder bounds, and large order

The Stirling series is divergent, but in the present problem its Borel transform is elementary. This gives more information than a formal expansion: it identifies the exact analytic function represented by the series and makes its remainder mechanism transparent.

5.1 Borel convention and Binet's remainder

For a formal inverse-power series

$$F(s) \sim \sum_{m\geq 0} \frac{a_m}{s^{m+1}}, \tag{43}$$

we use the formal Borel transform

$$\widehat{F}(t) = \sum_{m\geq 0} a_m \frac{t^m}{m!}. \tag{44}$$

Whenever the integral exists along the positive ray, its Laplace sum is

$$\mathscr{S}_0 F(s) = \int_0^\infty \mathrm{e}^{-st}\widehat{F}(t)\,\mathrm{d}t. \tag{45}$$

For $\Re z > 0$, define the exact Binet correction

$$\Omega(z) = \log\Gamma(z) - \left(z - \frac{1}{2}\right)\log z + z - \frac{1}{2}\log(2\pi). \tag{46}$$

Binet's integral can be written as

$$\Omega(z) = \int_0^\infty \mathrm{e}^{-zt}\left(\frac{\coth(t/2)}{2t} - \frac{1}{t^2}\right)\mathrm{d}t. \tag{47}$$

The apparent singularity at $t = 0$ is removable. Taylor expansion of the kernel gives Stirling's Bernoulli series, while exponential decay at infinity makes the integral convergent for $\Re z > 0$.

5.2 The two elementary Borel transforms

Define the exact uncentred and centred logarithmic corrections by

$$\mathcal{U}(x) = \log\mathcal{D}_r(x) - \frac{x}{2}(\log x - 1) - \frac{1}{2}\log x - C_r, \tag{48}$$

$$\mathcal{C}(s) = \log\mathcal{D}_r(s-1) - \frac{s}{2}(\log s - 1) - C_r. \tag{49}$$

They do not depend on r.

Theorem 5.1 (Exact Borel–Laplace representations). *For $x > 0$ and $s > 0$,*

$$\mathcal{U}(x) = \int_0^\infty \mathrm{e}^{-xt}\left(\frac{\coth t}{2t} - \frac{1}{2t^2}\right)\mathrm{d}t, \tag{50}$$

$$\mathcal{C}(s) = \int_0^\infty \mathrm{e}^{-st}\left(\frac{1}{2t\sinh t} - \frac{1}{2t^2}\right)\mathrm{d}t. \tag{51}$$

Their Borel transforms admit the Mittag–Leffler sums

$$\widehat{\mathcal{U}}(t) = \frac{\coth t}{2t} - \frac{1}{2t^2} = \sum_{m=1}^{\infty} \frac{1}{\pi^2 m^2 + t^2}, \tag{52}$$

$$\widehat{\mathcal{C}}(t) = \frac{1}{2t\sinh t} - \frac{1}{2t^2} = -\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{\pi^2 m^2 + t^2}. \tag{53}$$

Consequently Equations (22) *and* (26) *are Borel summable along the positive real direction, and their Borel sums are the exact gamma expressions.*

Proof. The uncentred correction is $\mathcal{U}(x) = \Omega(x/2)$. Substitute $t = 2u$ in Equation (47) to obtain Equation (50). From Equation (30), the centred correction is $\mathcal{C}(s) = \Omega(s) - \Omega(s/2)$. Subtracting the corresponding kernels and using

$$\coth(t/2) - \coth t = \frac{1}{\sinh t}$$

gives Equation (51). The classical Mittag–Leffler expansion

$$\frac{\coth t}{t} = \frac{1}{t^2} + 2\sum_{m\geq 1} \frac{1}{t^2 + \pi^2 m^2}$$

proves Equation (52). Splitting this sum into even and odd indices, or equivalently subtracting the scaled kernel, gives Equation (53). Both kernels are analytic on the positive integration ray and have at most exponential growth in sectors avoiding their poles, so the Laplace integrals are unambiguous there. □

Expanding the partial fractions at $t = 0$ immediately recovers

$$\widehat{\mathcal{U}}(t) = \sum_{k\geq 1} (-1)^{k-1} \frac{\zeta(2k)}{\pi^{2k}} t^{2k-2}, \tag{54}$$

$$\widehat{\mathcal{C}}(t) = \sum_{k\geq 1} (-1)^{k} \frac{\eta(2k)}{\pi^{2k}} t^{2k-2}, \tag{55}$$

which is equivalent to Equations (5) and (23) after Laplace integration.

5.3 Sharp alternating remainder bounds

Theorem 5.2 (First-omitted-term bounds). *For $x > 0$, $s > 0$, and $N \geq 0$, put*

$$U_N(x) = \sum_{k=1}^{N} \frac{\delta_k}{x^{2k-1}}, \tag{56}$$

$$C_N(s) = \sum_{k=1}^{N} \frac{\beta_k}{s^{2k-1}}. \tag{57}$$

Then

$$0 < (-1)^N \big(\mathcal{U}(x) - U_N(x)\big) < \frac{|\delta_{N+1}|}{x^{2N+1}}, \tag{58}$$

$$0 < (-1)^{N+1} \big(\mathcal{C}(s) - C_N(s)\big) < \frac{|\beta_{N+1}|}{s^{2N+1}}. \tag{59}$$

The convention is that the empty sum for $N = 0$ equals zero.

Proof. For $a > 0$,

$$\frac{1}{a^2 + t^2} = \sum_{j=0}^{N-1} \frac{(-1)^j t^{2j}}{a^{2j+2}} + \frac{(-1)^N t^{2N}}{a^{2N}(a^2 + t^2)}. \tag{60}$$

Insert this identity into Equation (52), sum over $a = \pi m$, and integrate. The remainder has sign $(-1)^N$; replacing $a^2 + t^2$ by a^2 gives the upper bound, and $\int_0^\infty \mathrm{e}^{-xt} t^{2N}\, \mathrm{d}t = (2N)!x^{-(2N+1)}$ yields Equation (58).

For the centred transform, the residual kernel is

$$(-1)^{N+1} t^{2N} S_N(t), \qquad S_N(t) = \sum_{m\geq 1} \frac{(-1)^{m-1}}{(\pi m)^{2N}\big((\pi m)^2 + t^2\big)}.$$

To see that $S_N(t) > 0$ and decreases with t^2, use

$$S_N(t) = \int_0^\infty \mathrm{e}^{-t^2 v} \sum_{m\geq 1} \frac{(-1)^{m-1}\mathrm{e}^{-\pi^2 m^2 v}}{(\pi m)^{2N}}\,\mathrm{d}v.$$

For every $v \geq 0$ the inner series is an alternating series with strictly decreasing positive terms, hence is positive. Therefore $0 < S_N(t) \leq S_N(0) = \eta(2N+2)/\pi^{2N+2}$. Laplace integration now gives Equation (59). □

These inequalities are particularly useful for inversion: they give a certified logarithmic residual without evaluating a huge double factorial. They also show directly that the centred series alternates with its first term negative.

5.4 Large-order growth and optimal truncation

The exact coefficient formulae imply

$$\delta_k = (-1)^{k-1}\frac{(2k-2)!}{\pi^{2k}}\left(1 + O(2^{-2k})\right), \tag{61}$$

$$\beta_k = (-1)^{k}\frac{(2k-2)!}{\pi^{2k}}\left(1 + O(2^{-2k})\right). \tag{62}$$

Thus the kth centred term is smallest when $2k \approx \pi s$. Stirling's formula then gives the least-term envelope

$$\min_k \left|\frac{\beta_k}{s^{2k-1}}\right| = O\left(s^{-1/2}\mathrm{e}^{-\pi s}\right). \tag{63}$$

The nearest Borel singularities are the simple poles $t = \pm \mathrm{i}\pi$; the full lattice is $t = \pm\mathrm{i}\pi m$, $m \geq 1$. This pole geometry simultaneously explains the factorial coefficient growth and the scale Equation (63).

Remark 5.3 (Exponentially improved interpretation). The optimal remainder on the positive real axis has the envelope $\mathrm{e}^{-\pi s}$, but the analytically continued hyperasymptotic sectors are not plain real exponentials. They are terminant-weighted contributions associated with singulants $\pm\mathrm{i}\pi m s$. Away from a Stokes direction the terminant multiplier itself is exponentially small. This distinction matters: appending $\mathrm{e}^{-\pi s}$ with an arbitrary constant to the divergent Bernoulli series is not a valid transseries completion. The exact Borel integral Equation (51) is the unambiguous positive-real completion.

6 Lambert normalization of the branchwise inverse

The centred phase is ideal for inversion because its dominant part has an exact Lambert-W inverse and its perturbation occupies only odd inverse powers. We first isolate the core, then solve the perturbation on a well-ordered q-grid.

6.1 Exact inversion of the core

Let

$$\Phi_0(s) = s(\log s - 1), \qquad s > 1. \tag{64}$$

For $R > 0$, the equation $\Phi_0(T) = 2R$ has an exact solution.

Lemma 6.1 (Lambert core)**.** *Define*

$$w = \mathrm{W}_0(2R/\mathrm{e}), \qquad T = \frac{2R}{w} = \mathrm{e}^{w+1}, \qquad \Lambda = w + 1 = \log T. \tag{65}$$

Then

$$T(\log T - 1) = 2R, \qquad \Phi_0'(T) = \Lambda, \qquad \frac{\mathrm{d}T}{\mathrm{d}R} = \frac{2}{\Lambda}. \tag{66}$$

Proof. Since $we^w = 2R/\mathrm{e}$, one has $T = 2R/w = \mathrm{e}^{w+1}$ and therefore $\log T - 1 = w$. Multiplication gives the first identity. The other two follow by differentiation. □

For the parity branch r, take $R = R_r = \log y - C_r$. The leading inverse is $x = T_r - 1$. Retaining W_0 exactly packages infinitely many outer logarithmic terms into one analytic quantity; this is both more accurate and algebraically cleaner than expanding W_0 at the outset.

6.2 The exact normalized formal equation

Multiply Equation (26) by 2 and use $\alpha_k = 2\beta_k$. Formally, solving $\mathcal{D}_r(s-1) = y$ is equivalent to

$$2R_r = s(\log s - 1) + \sum_{k\geq 1} \frac{\alpha_k}{s^{2k-1}}. \tag{67}$$

Let $T = T_r$, $\Lambda = \Lambda_r$, $q = T^{-1}$, and write

$$s = T(1+E), \qquad E = O(q^2). \tag{68}$$

Subtracting the core equation and dividing by T gives

$$\Lambda E + \varphi(E) + \sum_{k\geq 1} \alpha_k q^{2k}(1+E)^{-(2k-1)} = 0, \tag{69}$$

where

$$\varphi(u) = (1+u)\log(1+u) - u = \sum_{m\geq 2} \frac{(-1)^m}{m(m-1)} u^m. \tag{70}$$

The support is $2\mathbb{N}$: if E contains only even powers of q, so does the right-hand side, and the linear coefficient Λ is invertible in the coefficient field $\mathbb{Q}(\Lambda)$.

7 All-orders parity-resolved inverse transseries

7.1 Triangular recurrence and analytic remainder

Theorem 7.1 (Complete branchwise inverse expansion)**.** *Let $r \in \{0,1\}$ and define R_r, w_r, T_r, Λ_r by Equation (7). For every fixed $N \geq 0$,*

$$\mathcal{D}_r^{-1}(y) = T_r - 1 + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda_r)}{T_r^{2n-1}} + O\left(\frac{1}{T_r^{2N+1}\Lambda_r}\right), \qquad y \to +\infty. \tag{71}$$

The coefficients are uniquely determined as follows. Put

$$E_{n-1}(q) = \sum_{j=1}^{n-1} e_j(\Lambda) q^{2j}. \tag{72}$$

Then

$$e_n(\Lambda) = -\frac{1}{\Lambda}\left[q^{2n}\right]\left[\varphi(E_{n-1}) + \sum_{k=1}^{n} \alpha_k q^{2k}(1+E_{n-1})^{-(2k-1)}\right], \tag{73}$$

with

$$d_{2n-1}(\Lambda) = e_n(\Lambda). \tag{74}$$

Furthermore,

$$d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} d_{2n-1} \leq 2n-1, \tag{75}$$

and

$$d_{2n-1}(\Lambda) = -\frac{\alpha_n}{\Lambda} + O(\Lambda^{-2}). \tag{76}$$

Proof. Seek a solution of Equation (69) in the form $E = \sum_{n\geq 1} e_n q^{2n}$. At order q^{2n}, the only term containing the unknown e_n is $\Lambda e_n q^{2n}$. The nonlinear series $\varphi(E)$ starts with $E^2/2$, and every term in the Bernoulli sector has positive q-valuation. Consequently the remaining coefficient of q^{2n} depends only on $e_1, \dots, e_{n-1}$, and solving the coefficient equation gives Equation (73). This proves formal existence and uniqueness by induction.

The same induction proves Equation (75). The coefficient inside the brackets is a polynomial in Λ^{-1} of degree at most $2n-2$; division by Λ gives the claimed bound. The only contribution at q^{2n} that is independent of lower e_j is $\alpha_n q^{2n}$, proving Equation (76).

For the analytic statement, truncate the exact centred Borel sum after the term $\beta_{N+1} s^{-(2N+1)}$. By Theorem 5.2, the omitted forward logarithmic residual is $O(s^{-(2N+3)})$. Substituting the formal inverse through $e_N q^{2N}$ cancels the doubled phase through order q^{2N} and leaves a residual $O(T^{-(2N+1)})$. The derivative of the doubled phase is

$$\log s - 2\sum_{k\geq 1}(2k-1)\beta_k s^{-2k} = \Lambda + O(T^{-2}). \tag{77}$$

For large T it is bounded below by a fixed positive multiple of Λ. The mean-value theorem therefore transfers the forward residual to an inverse error $O(T^{-(2N+1)}/\Lambda)$. □

Remark 7.2 (Uniformity in parity). The implied constants in Equation (71) can be chosen uniformly for $r = 0, 1$. The coefficient recurrence contains no parity label; r enters only through the bounded constants C_r in R_r.

7.2 Explicit inverse blocks

Successive use of Equation (73) gives

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \tag{78}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \tag{79}$$

$$d_5(\Lambda) = \frac{1}{45360\Lambda^5}\left(2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105\right), \tag{80}$$

$$\begin{aligned} d_7(\Lambda) = -\frac{1}{5443200\Lambda^7}\big(&822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 \\ &+ 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625\big), \end{aligned} \tag{81}$$

$$\begin{aligned} d_9(\Lambda) = \frac{1}{359251200\Lambda^9}\big(&309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 \\ &+ 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 \\ &+ 242550\Lambda + 40425\big). \end{aligned} \tag{82}$$

Thus the first three correction blocks are

$$\begin{aligned} \mathcal{D}_r^{-1}(y) = T_r - 1 + \frac{1}{6T_r\Lambda_r} - \frac{14\Lambda_r^2 + 10\Lambda_r + 5}{360T_r^3\Lambda_r^3} \\ + \frac{2232\Lambda_r^4 + 1176\Lambda_r^3 + 714\Lambda_r^2 + 350\Lambda_r + 105}{45360T_r^5\Lambda_r^5} + O\left(\frac{1}{T_r^7\Lambda_r}\right). \end{aligned} \tag{83}$$

For completeness, the next block is

$$\begin{aligned} d_{11}(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}}&\big(46195687238400\Lambda^{10} + 8854370366400\Lambda^9 \\ &+ 2171643318000\Lambda^8 + 605515022112\Lambda^7 + 212869408752\Lambda^6 + 84136852800\Lambda^5 \\ &+ 35589153600\Lambda^4 + 14234019800\Lambda^3 + 4868113250\Lambda^2 + 1213962750\Lambda \\ &+ 165540375\big). \end{aligned} \tag{84}$$

The rapidly growing numerators are an argument for using the general formulae rather than storing a table.

7.3 Low-order derivation by hand

At order q^2, Equation (69) gives $\Lambda e_1 + \alpha_1 = 0$. Since $\alpha_1 = -1/6$,

$$e_1 = \frac{1}{6\Lambda}.$$

At order q^4 one obtains

$$\Lambda e_2 + \frac{1}{2}e_1^2 - \alpha_1 e_1 + \alpha_2 = 0. \tag{85}$$

Substitution of $\alpha_1 = -1/6$, $\alpha_2 = 7/180$ yields Equation (79). At every subsequent order the calculation is finite; the number of additive partitions grows, but no new analytic idea is required.

8 A finite closed formula for every inverse coefficient

The triangular recurrence is optimal for computation, but it conceals the combinatorial content. Lagrange–Bürmann inversion converts it into a finite, non-recursive expression involving two coefficient arrays: one built from the Bernoulli data and one from the universal nonlinear kernel φ.

8.1 Bernoulli convolution and kernel arrays

Define

$$A(z) = \sum_{k\geq 1} \alpha_k z^k, \qquad A_{n,p} = [z^n]\, A(z)^p. \tag{86}$$

By the defining identity for partial Bell polynomials,

$$A_{n,p} = \frac{p!}{n!}\mathrm{B}_{n,p}\big(1!\alpha_1, 2!\alpha_2,\ \ldots, (n-p+1)!\alpha_{n-p+1}\big). \tag{87}$$

For integers n, j, p define

$$K_{n;j,p} = \big[u^{j-1}\big]\, \varphi(u)^{j-p}(1+u)^{-(2n-p)}. \tag{88}$$

If $p > j$, or if $2(j-p) > j-1$, this coefficient is zero.

To express $K_{n;j,p}$ itself through Bell polynomials, put

$$\nu_1 = 0, \qquad \nu_m = (-1)^m(m-2)! \quad (m \geq 2), \tag{89}$$

so that $\varphi(u) = \sum_{m\geq 1} \nu_m u^m/m!$. With $r = j-p$ and the conventions $\mathrm{B}_{0,0} = 1$ and $\mathrm{B}_{h,0} = 0$ for $h > 0$, one has the finite formula

$$\begin{aligned} K_{n;j,p} = \sum_{h=2r}^{j-1} \frac{r!}{h!}&\mathrm{B}_{h,r}(\nu_1, \ldots, \nu_{h-r+1})(-1)^{j-1-h} \\ &\times \binom{2n-p+j-2-h}{j-1-h}. \end{aligned} \tag{90}$$

When $r = 0$, only the $h = 0$ term is retained.

Theorem 8.1 (Closed coefficient formula)**.** *Write*

$$d_{2n-1}(\Lambda) = \sum_{j=1}^{2n-1} c_{n,j}\Lambda^{-j}. \tag{91}$$

Then, for $1 \le j \le 2n-1$,

$$\boxed{c_{n,j} = \frac{(-1)^j}{j} \sum_{p=\lceil (j+1)/2 \rceil}^{\min(j,n)} \binom{j}{p} A_{n,p} K_{n;j,p}.} \tag{92}$$

Equations (87) and (90) make the right-hand side a completely finite expression in Bernoulli numbers, binomial coefficients, and partial Bell polynomials.

Proof. Introduce an auxiliary homotopy parameter ζ and rewrite Equation (69) as

$$E = -\frac{\zeta}{\Lambda} H(q,E), \qquad H(q,u) = \varphi(u) + \sum_{k\ge 1} \alpha_k q^{2k} (1+u)^{-(2k-1)}. \tag{93}$$

The formal Lagrange–Bürmann formula gives

$$E(q,\zeta) = \sum_{j\ge 1} \frac{(-1)^j \zeta^j}{j!\Lambda^j} \frac{\mathrm{d}^{j-1}}{\mathrm{d}u^{j-1}} H(q,u)^j \bigg|_{u=0}. \tag{94}$$

Set $\zeta = 1$ and extract q^{2n}. In H^j, choose p of the j factors from the Bernoulli sector and the other $j-p$ from φ. The binomial choice contributes $\binom{j}{p}$. If the selected Bernoulli indices sum to n, then

$$\prod_{\ell=1}^{p} (1+u)^{-(2k_\ell - 1)} = (1+u)^{-(2n-p)},$$

and their coefficient sum is $A_{n,p}$. The remaining coefficient of u^{j-1} is $K_{n;j,p}$. Since differentiation contributes $(j-1)!$, the factor $1/j!$ becomes $1/j$, proving Equation (92).

The factor φ^{j-p} starts in degree $2(j-p)$, so a nonzero u^{j-1} coefficient requires $p \ge \lceil (j+1)/2 \rceil$. As $p \le n$, this also forces $j \le 2n-1$, proving directly that the sum in Equation (91) terminates. Finally, expanding φ^r by Equation (33) and multiplying by the binomial series for $(1+u)^{-(2n-p)}$ gives Equation (90). ☐

Coefficient calculus

There are now two all-orders algorithms.

- The recurrence in Equation (73) generates complete blocks $d_1, d_3, \dots, d_{2N-1}$ with aggressive truncation in q and is the fastest route in a computer algebra system.
- The closed formula in Equation (92) computes any isolated scalar coefficient $c_{n,j}$ without constructing lower inverse blocks. It is the better formula for sign, support, denominator, and boundary analyses.

Both use exactly the Bell-polynomial/Faà di Bruno coefficient machinery of combinatorial coefficient calculus.

8.2 Boundary coefficients

Two edges of the triangular array $c_{n,j}$ simplify immediately.

Corollary 8.2 (First and deepest inverse-logarithmic coefficients)**.** *For* $n \ge 1$,

$$c_{n,1} = -\alpha_n, \tag{95}$$

$$c_{n,2n-1} = -\frac{1}{2^{n-1}(2n-1)} \binom{2n-1}{n} \alpha_1^n. \tag{96}$$

Proof. For $j=1$, only $p=1$ contributes, $K_{n;1,1}=1$, and $A_{n,1}=\alpha_n$. For $j=2n-1$, the support restriction forces $p=n$. Then $A_{n,n}=\alpha_1^n$, and the required coefficient is the lowest term of $\varphi(u)^{n-1}$, namely $[u^{2n-2}](u^2/2)^{n-1} = 2^{1-n}$. Substitution in Equation (92) proves the result. ☐

8.3 Global alternating sign law

The signs visible in Equations (78) to (82) persist in every coefficient of every block.

Theorem 8.3 (Coefficientwise sign theorem)**.** *For every $n \geq 1$,*

$$(-1)^{n+1} d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}_{>0}[\Lambda^{-1}]. \tag{97}$$

In particular, every nonzero scalar coefficient $c_{n,j}$ has sign $(-1)^{n+1}$.

Proof. By Equation (6), write

$$\alpha_k = (-1)^k a_k^+, \qquad a_k^+ = \frac{2(2k-2)!\eta(2k)}{\pi^{2k}} > 0. \tag{98}$$

Set $z = -q^2$ and $E = -P$. Since

$$\varphi(-P) = \sum_{m\geq 2} \frac{P^m}{m(m-1)}, \tag{99}$$

all coefficients are positive, and the normalized equation becomes

$$\Lambda P = \sum_{m\geq 2} \frac{P^m}{m(m-1)} + \sum_{k\geq 1} a_k^+ z^k (1-P)^{-(2k-1)}. \tag{100}$$

Starting from $P = 0$ and solving coefficient by coefficient first gives $P = \sum_{n\geq 1} p_n(\Lambda^{-1})z^n$ with nonnegative rational coefficients. Strict positivity at every power Λ^{-j}, $1 \leq j \leq 2n-1$, follows constructively. For $1 \leq j \leq n$, take the forcing term with index $n-j+1$ and the P^{j-1} term in its positive binomial expansion, using the leading $a_1^+ z/\Lambda$ contribution in every copy of P. For $n < j \leq 2n-1$, use a positive rooted-tree contribution with n leaves and $j-n$ internal vertices from the nonlinear series $\sum_{m\geq 2} P^m/(m(m-1))$; such a tree exists for every $1 \leq j-n \leq n-1$. Hence $p_n \in \Lambda^{-1}\mathbb{Q}_{>0}[\Lambda^{-1}]$. Returning to q,

$$E(q) = -P(-q^2) = \sum_{n\geq 1} (-1)^{n+1} p_n q^{2n}.$$

Thus $d_{2n-1} = (-1)^{n+1}p_n$, proving the theorem. $\square$

8.4 Scaling relation with the inverse-Gamma blocks

Let $a_k^\Gamma = B_{2k}(1/2)/(2k(2k-1))$ be the half-shifted logarithmic coefficients used for inverse Gamma. Since

$$\alpha_k = 4^k a_k^\Gamma, \tag{101}$$

the normalized double-factorial equation is obtained from the normalized inverse-Gamma equation by replacing q with $2q$, while leaving Λ formal. Therefore

$$d_{2n-1}^{!!}(\Lambda) = 4^n d_{2n-1}^\Gamma(\Lambda). \tag{102}$$

This identity is an independent all-orders check on both recurrences. It is a coefficient-polynomial identity; the numerical Lambert variables used in the two inverse problems need not coincide.

9 Outer iterated-logarithmic expansion

The Lambert form is the natural compact transseries. When an expansion using only elementary iterated logarithms is required, W can itself be expanded. The resulting coefficient calculus is again governed by classical combinatorial numbers.

9.1 Unsigned-Stirling formula for Lambert W

Set

$$Z = \frac{2R}{\mathrm{e}}, \qquad L_1 = \log Z, \qquad L_2 = \log L_1. \tag{103}$$

For $Z \to +\infty$, the complete Poincaré expansion is

$$\mathrm{W}_0(Z) \sim L_1 - L_2 + \sum_{a\ge 0}\sum_{b\ge 1} \frac{(-1)^a}{b!} \begin{bmatrix} a+b \\ a+1 \end{bmatrix} \frac{L_2^b}{L_1^{a+b}}, \tag{104}$$

where $\left[{n \atop k}\right]$ is the unsigned Stirling number of the first kind [5, 4]. This double sum is understood by increasing total power $a+b$.

Write

$$\frac{\mathrm{W}_0(Z)}{L_1} = 1 + \sum_{m\ge 1} \frac{u_m(L_2)}{L_1^m}. \tag{105}$$

Then

$$u_1(L_2) = -L_2, \tag{106}$$

$$u_m(L_2) = \sum_{b=1}^{m-1} \frac{(-1)^{m-1-b}}{b!} \begin{bmatrix} m-1 \\ m-b \end{bmatrix} L_2^b, \qquad m \ge 2. \tag{107}$$

Thus every outer coefficient is an explicit polynomial in L_2 with integer Stirling data.

9.2 Reciprocal coefficients and the core

Define $\tau_m(L_2)$ by

$$\frac{1}{\mathrm{W}_0(Z)} \sim \frac{1}{L_1} \sum_{m\ge 0} \frac{\tau_m(L_2)}{L_1^m}. \tag{108}$$

Then

$$\tau_0 = 1, \qquad \tau_m = -\sum_{j=1}^{m} u_j \tau_{m-j}. \tag{109}$$

A non-recursive partition form is

$$\tau_m = \sum_{\substack{k_1,\dots,k_m\ge 0 \\ \sum_{j=1}^m jk_j = m}} (-1)^K \frac{K!}{k_1!\cdots k_m!} \prod_{j=1}^m u_j^{k_j}, \qquad K = \sum_{j=1}^m k_j. \tag{110}$$

The first polynomials are

$$\begin{aligned}
\tau_0 &= 1, \\
\tau_1 &= L_2, \\
\tau_2 &= L_2(L_2-1), \\
\tau_3 &= \frac{1}{2} L_2(2L_2^2 - 5L_2 + 2), \\
\tau_4 &= \frac{1}{6} L_2(6L_2^3 - 26L_2^2 + 27L_2 - 6), \\
\tau_5 &= \frac{1}{12} L_2(12L_2^4 - 77L_2^3 + 142L_2^2 - 84L_2 + 12).
\end{aligned} \tag{111}$$

Since $T = 2R/W_0(Z)$,

$$T \sim \frac{2R}{L_1}\left[1 + \frac{L_2}{L_1} + \frac{L_2(L_2-1)}{L_1^2} + \frac{L_2(2L_2^2-5L_2+2)}{2L_1^3} + \frac{L_2(6L_2^3-26L_2^2+27L_2-6)}{6L_1^4} + O\left(\frac{L_2^5}{L_1^5}\right)\right]. \tag{112}$$

Equations (107) and (110) give every omitted coefficient.

9.3 All coefficients after expanding the inverse blocks

The compact complete branchwise transseries can be written directly in R and $w = W_0(2R/\mathrm{e})$ as

$$\boxed{\mathcal{D}_r^{-1}(y) \sim \frac{2R_r}{w_r} - 1 + \sum_{n\geq 1}\sum_{j=1}^{2n-1} c_{n,j}\left(\frac{w_r}{2R_r}\right)^{2n-1}(w_r+1)^{-j}.} \tag{113}$$

Together, Equations (92) and (104) are already a general formula for every coefficient of the fully elementary expansion. For completeness we spell out the final composition step.

Put

$$v_1 = 1 + u_1 = 1 - L_2, \qquad v_m = u_m \quad (m \geq 2), \tag{114}$$

so that

$$w + 1 = L_1\left(1 + \sum_{m\geq 1}\frac{v_m}{L_1^m}\right). \tag{115}$$

For positive integers p, j, define

$$(1 + \textstyle\sum_{m\geq 1} u_m z^m)^p = \displaystyle\sum_{m\geq 0}\sigma_m^{(p)} z^m, \tag{116}$$

$$(1 + \textstyle\sum_{m\geq 1} v_m z^m)^{-j} = \displaystyle\sum_{m\geq 0}\rho_m^{(j)} z^m. \tag{117}$$

Their finite partition formulae are

$$\sigma_m^{(p)} = \sum_{\substack{k_1,\dots,k_m\geq 0\\ \sum hk_h = m}}\binom{p}{K}\frac{K!}{\prod_h k_h!}\prod_{h=1}^m u_h^{k_h}, \tag{118}$$

$$\rho_m^{(j)} = \sum_{\substack{k_1,\dots,k_m\geq 0\\ \sum hk_h = m}}(-1)^K\binom{j+K-1}{K}\frac{K!}{\prod_h k_h!}\prod_{h=1}^m v_h^{k_h}, \tag{119}$$

where $K = \sum_h k_h$. It follows that

$$w^p(w+1)^{-j} \sim L_1^{p-j}\sum_{m\geq 0}\frac{1}{L_1^m}\sum_{a+b=m}\sigma_a^{(p)}\rho_b^{(j)}. \tag{120}$$

Substituting $p = 2n-1$ in Equation (120) and then using Equation (113) gives a finite combinatorial formula for any prescribed coefficient at any prescribed triple of levels (R^{-1}, L_1^{-1}, L_2).

9.4 Ordering of the scales

For every fixed M,

$$\frac{T}{L_1^M} \longrightarrow \infty. \tag{121}$$

Therefore the fixed centring shift and the first Bernoulli blocks obey

$$\frac{T}{L_1^M} \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \frac{1}{T^5\Lambda} \gg \cdots . \tag{122}$$

This is the reason for retaining -1 explicitly in Equation (71). It is invisible to every finite outer Lambert expansion, yet dominates the complete inverse-power grid.

10 The inverse of the periodic OEIS continuation

The one-function continuation Equation (14) is useful, but its inverse cannot be obtained by merely inserting a parity-averaged constant into the branchwise formula. The periodic factor creates a new sector before the Bernoulli grid.

10.1 Exact displacement equation

Let

$$R_* = \log y - C_*, \qquad w = \mathrm{W}_0(2R_*/\mathrm{e}), \qquad T = \frac{2R_*}{w}, \qquad \Lambda = w+1, \qquad q = T^{-1}, \qquad \theta = \pi T. \tag{123}$$

Write the inverse in centred form as

$$\mathfrak{D}_+^{-1}(y) = T - 1 + \Delta. \tag{124}$$

Using the exact identity in Equation (16), the displacement satisfies

$$\Lambda\Delta + q^{-1}\varphi(q\Delta) - 2a\cos(\theta + \pi\Delta) + 2\mathcal{C}(T+\Delta) = 0. \tag{125}$$

The second term begins with $q\Delta^2/2$; the Borel correction begins with $-q/6$. In contrast, the cosine term is $O(1)$. Thus $\Delta = O(\Lambda^{-1})$, not $O(q\Lambda^{-1})$.

10.2 The complete zero-grid log-periodic sector

At $q = 0$, Equation (125) reduces to

$$\Lambda\Delta_0 = 2a\cos(\theta + \pi\Delta_0). \tag{126}$$

For $\Lambda > 2\pi a$, the right-hand side divided by Λ is a contraction on the real line, so Equation (126) has a unique real solution. Its complete inverse-logarithmic expansion has closed Fourier coefficients.

Theorem 10.1 (Closed log-periodic sector). *Let $\lambda = \Lambda^{-1}$. The solution of Equation (126) has the expansion*

$$\Delta_0(\theta, \lambda) = \sum_{m\ge 1} p_m(\theta)\lambda^m, \tag{127}$$

where

$$p_m(\theta) = \frac{(2a)^m}{m!}\,\frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}}\cos^m(\theta + \pi u)\bigg|_{u=0} \tag{128}$$

$$= \frac{a^m}{m!}\sum_{j=0}^{m}\binom{m}{j}\big(\mathrm{i}\pi(m-2j)\big)^{m-1}\mathrm{e}^{\mathrm{i}(m-2j)\theta}. \tag{129}$$

The second expression is real despite its complex notation. The first terms are

$$\Delta_0 = \frac{2a\cos\theta}{\Lambda} - \frac{4\pi a^2\cos\theta\sin\theta}{\Lambda^2} + \frac{4\pi^2 a^3\cos\theta(2-3\cos^2\theta)}{\Lambda^3} + O(\Lambda^{-4}). \tag{130}$$

Proof. Equation (126) is $\Delta_0 = 2a\lambda\cos(\theta+\pi\Delta_0)$. Lagrange–Bürmann inversion in the parameter λ gives Equation (128). Expanding $\cos^m v = 2^{-m}\sum_{j=0}^{m}\binom{m}{j}\mathrm{e}^{\mathrm{i}(m-2j)v}$ and differentiating gives Equation (129). Direct evaluation for $m = 1, 2, 3$ yields Equation (130). □

The formula in Equation (129) exhibits the precise harmonic content: at inverse-logarithmic order m, only Fourier modes $m, m-2, \ldots, -m$ occur. Thus the periodic sector forms a finitely generated coefficient algebra at every order, even though it lies outside ordinary polynomial-logarithmic transseries.

10.3 Coupling the periodic sector to all inverse powers

Replace the exact $\mathcal{C}$ in Equation (125) by its Borel-summable asymptotic series and define the formal function

$$\mathcal{F}(q,u) = \Lambda u + \sum_{m\geq 2}\frac{(-1)^m}{m(m-1)}u^m q^{m-1} - 2a\cos(\theta+\pi u) + 2\sum_{k\geq 1}\beta_k q^{2k-1}(1+qu)^{-(2k-1)}. \tag{131}$$

Seek

$$\Delta = \Delta_0 + \sum_{n\geq 1}\Delta_n q^n, \tag{132}$$

where each Δ_n is a formal series in Λ^{-1} with trigonometric polynomial coefficients. Put

$$M = \Lambda + 2\pi a\sin(\theta+\pi\Delta_0). \tag{133}$$

The leading term of M is Λ, so it is invertible in this coefficient field.

Theorem 10.2 (Triangular periodic-grid recurrence). *For $n \geq 1$, let*

$$\Delta_{<n}(q) = \Delta_0 + \sum_{j=1}^{n-1}\Delta_j q^j. \tag{134}$$

Then the unique formal solution of Equation (131) is generated by

$$\boxed{\Delta_n = -\frac{1}{M}\,[q^n]\,\mathcal{F}\big(q,\Delta_{<n}(q)\big).} \tag{135}$$

The first coefficient is

$$\Delta_1 = \frac{1/6 - \Delta_0^2/2}{M}. \tag{136}$$

The second is

$$\Delta_2 = -\frac{1}{M}\left[a\pi^2\cos(\theta+\pi\Delta_0)\Delta_1^2 + \Delta_0\Delta_1 + \frac{\Delta_0 - \Delta_0^3}{6}\right]. \tag{137}$$

Proof. The coefficient of a new $\Delta_n q^n$ in the q^n part of $\mathcal{F}(q,\Delta)$ is the derivative of the zero-grid function $\Lambda u - 2a\cos(\theta+\pi u)$ at $u = \Delta_0$, namely M. Every other contribution involves only lower Δ_j. Solving the coefficient equation gives Equation (135).

At order q, the core kernel contributes $\Delta_0^2/2$, and $2\beta_1 = -1/6$ supplies the Bernoulli contribution, proving Equation (136). At order q^2, Taylor expansion of the zero-grid function, the $u^2q/2 - u^3q^2/6$ core kernel, and the expansion of $2\beta_1 q(1+qu)^{-1}$ give Equation (137). □

Corollary 10.3 (Asymptotic meaning of the periodic grid)**.** *Fix nonnegative integers N and M. Truncate Equation (132) after q^N, and in each coefficient retain the inverse-logarithmic expansion through Λ^{-M}. If the centred forward series is retained far enough to determine these coefficients, substitution into Equation (125) leaves a residual that is smaller in the lexicographic scale*

$$1, \Lambda^{-1}, \Lambda^{-2}, \dots; \quad q, q\Lambda^{-1}, q\Lambda^{-2}, \dots; \quad q^2, \dots.$$

Since the exact derivative with respect to Δ equals $M + O(q)$ and $M \sim \Lambda$, the residual transfers to an inverse error after division by Λ. The assertion is uniform in the phase $\theta = \pi T$, because all trigonometric coefficients and their derivatives are bounded.

Proof. The recurrence cancels the formal residual successively in q. At each fixed q-order, Lagrange inversion cancels it successively in Λ^{-1}. The exact Borel remainder estimate Theorem 5.2 controls the replacement of $\mathcal{C}$ by its finite Bernoulli expansion. Finally, the implicit derivative is bounded below by a positive multiple of Λ for large T, uniformly in θ, so the mean-value argument used in Theorem 7.1 applies. □

Summary

The eventual inverse of the OEIS continuation is

$$\mathfrak{D}_+^{-1}(y) \sim T - 1 + \Delta_0(\pi T, \Lambda^{-1}) + \sum_{n\geq 1} \Delta_n(\pi T, \Lambda^{-1}) T^{-n}. \tag{138}$$

The coefficient Δ_0 is given explicitly by Equations (128) and (129); all finer coefficients are generated by Equation (135). This is a log-periodic transseries over the monomial group generated by Λ^{-1}, T^{-1}, and $\mathrm{e}^{\mathrm{i}\pi T}$.

10.4 How the parity branches resum the periodic sector

At an integer n, $s = n + 1$ and

$$C_* - a\cos(\pi s) = \begin{cases} C_0, & n \text{ even}, \\ C_1, & n \text{ odd}. \end{cases} \tag{139}$$

The branchwise normalization absorbs this entire constant before the Lambert inversion. Hence the large $O(\Lambda^{-1})$ periodic displacement disappears, and the first remaining correction is the much smaller $1/(6T\Lambda)$. In this precise sense the two parity-resolved transseries are subsequence resummations of the periodic inverse: they build the parity phase into the core instead of expanding it perturbatively.

11 Divergence and beyond-all-orders inverse accuracy

The branchwise inverse series is Poincaré asymptotic, not convergent. An exact large-order statement follows immediately from the shallow edge of the coefficient triangle:

$$[\Lambda^{-1}]\, d_{2n-1}(\Lambda) = (-1)^{n+1} \frac{2(2n-2)!\,\eta(2n)}{\pi^{2n}} = (-1)^{n+1} \frac{2(2n-2)!}{\pi^{2n}} \left(1 + O(2^{-2n})\right). \tag{140}$$

At this fixed inverse-logarithmic depth, the term $[\Lambda^{-1}] d_{2n-1}/T^{2n-1}$ is smallest near $2n \approx \pi T$. The deeper powers of Λ^{-1} are not uniform in that diagonal limit, so Equation (140) alone should not be used as a proof of optimal truncation for the full block polynomial.

A rigorous envelope follows instead from the exact forward remainder. Divide the optimally truncated centred logarithmic residual Equation (63) by the inverse slope $\frac{1}{2}\Lambda + O(T^{-2})$. This gives

$$\mathrm{error}_{\mathrm{opt}} = O\left(\frac{\mathrm{e}^{-\pi T}}{\sqrt{T}\,\Lambda}\right). \tag{141}$$

Thus the exponential envelope is established without assuming any unproved uniform large-n estimate for the complete polynomials d_{2n-1}.

Proposition 11.1 (Transport of a forward beyond-all-orders sector)**.** *Suppose a chosen exponentially improved representation of the centred forward phase retains the algebraic terms $\beta_1, \ldots, \beta_K$ and has an additional residual $\mathcal{E}(s)$. Let s_{alg} solve the correspondingly truncated algebraic inverse problem. Then the leading induced inverse correction is*

$$\Delta_{\mathcal{E}} = -\frac{2\mathcal{E}(s_{\text{alg}})}{\log s_{\text{alg}} - 2\sum_{k=1}^{K}(2k-1)\beta_k s_{\text{alg}}^{-2k}} + O\left(\frac{\sup |F''_{\text{alg}}|\, |\mathcal{E}|^2}{\Lambda^3} + \frac{|\mathcal{E}\,\mathcal{E}'|}{\Lambda^2}\right), \tag{142}$$

where F_{alg} is the doubled algebraic phase and the supremum is taken between the algebraic and exact inverse points. The estimate assumes the displayed denominator is comparable to Λ and $|\mathcal{E}'| = o(\Lambda)$.

Proof. Let $F(s)$ denote the exact doubled forward phase minus $2R$, and let F_{alg} denote the same expression with $\mathcal{E}$ omitted. By construction, $F_{\text{alg}}(s_{\text{alg}}) = 0$ and $F(s_{\text{alg}}) = 2\mathcal{E}(s_{\text{alg}})$. Taylor expansion $0 = F(s_{\text{alg}} + \Delta) = F(s_{\text{alg}}) + F'(s_{\text{alg}})\Delta + O(F''\Delta^2)$ gives the first Newton correction. Replacing $F'(s_{\text{alg}}) = F'_{\text{alg}}(s_{\text{alg}}) + 2\mathcal{E}'(s_{\text{alg}})$ by F'_{alg} and iterating once produces the two displayed remainder terms and proves Equation (142). □

This proposition is deliberately stated in terms of a supplied forward sector. The Borel poles at $\pm \mathrm{i}\pi m$ generate terminant functions whose form and Stokes multipliers depend on the sector of analytic continuation. The transport law in Equation (142) carries any such rigorously normalized forward completion without inventing a spurious exponential term.

12 Complex inverse sheets

Although the principal application is the large positive real inverse, the Lambert normalization also labels complex asymptotic sheets. Fix a branch of $\operatorname{Log} y$ and an integer logarithmic winding number m. Put

$$R_{r,m} = \operatorname{Log} y + 2\pi \mathrm{i} m - C_r. \tag{143}$$

For a Lambert branch $k \in \mathbb{Z}$, define

$$w_{r,m,k} = \mathrm{W}_k(2R_{r,m}/\mathrm{e}), \qquad T_{r,m,k} = \exp(w_{r,m,k} + 1) = \frac{2R_{r,m}}{w_{r,m,k}}, \qquad \Lambda_{r,m,k} = w_{r,m,k} + 1. \tag{144}$$

Whenever $T_{r,m,k}$ lies in a sector in which the gamma Stirling expansion is uniform and avoids the poles of the forward branch, the same coefficient polynomials give the local formal sheet

$$x_{r,m,k}(y) \sim T_{r,m,k} - 1 + \sum_{n \geq 1} \frac{d_{2n-1}(\Lambda_{r,m,k})}{T_{r,m,k}^{2n-1}}. \tag{145}$$

The pair (m, k) is an asymptotic label, not a global classification: distinct labels can meet after analytic continuation around gamma poles or logarithmic cuts. The expansion in Equation (145) nevertheless gives a convenient local atlas at large modulus.

13 Numerical evaluation and certification

13.1 Stable evaluation from a logarithmic target

Directly forming $y = n!!$ quickly overflows fixed-precision arithmetic. The inverse should therefore accept $\ell = \log y$ as its primary input.

Algorithm 13.1 (Parity-resolved inverse double factorial)**.** Given $\ell = \log y$, parity $r \in \{0,1\}$, and a desired number N of algebraic correction blocks:

```
C := (log(pi))/2              if r = 0
     (log(2))/2               if r = 1
R := ell - C
w := LambertW(0, 2*R/e)
T := 2*R/w
Lambda := w + 1
x := T - 1
for n := 1,...,N:
    x := x + d[2*n-1](Lambda) / T^(2*n-1)
return x
```

The polynomials d_{2n-1} may be precomputed by Equation (73), or individual scalar coefficients may be generated by Equation (92).

For full working precision, use the transseries approximation as a starting point for Newton's method applied to the logarithmic residual

$$F_r(x) = \log \kappa_r + \frac{x}{2}\log 2 + \log\Gamma\Big(\frac{x}{2}+1\Big) - \ell. \tag{146}$$

Its derivative is the positive quantity in Equation (13), so the Newton update is

$$x_{\text{new}} = x - \frac{F_r(x)}{\frac{1}{2}\left(\log 2 + \psi(x/2+1)\right)}. \tag{147}$$

A few asymptotic blocks normally place the initial value well inside the quadratic convergence region.

For the OEIS interpolation, add $\frac{1}{4}(1-\cos\pi x)\log(2/\pi)$ to $F_0(x)$ and its derivative to the Newton denominator. The transseries in Equation (138) gives the appropriate initial approximation.

13.2 A posteriori inverse certificate

Let $\widetilde{x}$ be any approximation and let $\rho = F_r(\widetilde{x})$. If an interval I containing both $\widetilde{x}$ and the exact inverse has

$$m_I = \inf_{x\in I} F_r'(x) > 0, \tag{148}$$

then the mean-value theorem gives

$$|\mathcal{D}_r^{-1}(y) - \widetilde{x}| \le \frac{|\rho|}{m_I}. \tag{149}$$

Because F_r' is increasing, a convenient lower bound is its value at the left endpoint of I. Combining Equation (149) with the parity floor in Equation (18) certifies the exact integer generalized inverse whenever the resulting interval does not cross a same-parity integer.

13.3 Numerical validation

To test the expansion without root-finding ambiguity, choose an exact index n, set $\ell = \log \mathcal{D}_r(n)$, reconstruct the Lambert variables from ℓ, and compare each truncation with n. The following values were computed with 80-decimal arithmetic. Each entry is approximation minus exact index; E_0 uses only $T-1$, and E_j successively includes $d_1, d_3, \ldots, d_{2j-1}$.

The parity dependence is confined to the Lambert core. Once the proper constant C_r is used, the same sequence of coefficient polynomials produces nearly identical convergence patterns on the two subsequences.

Table 2: Errors for the even branch $r = 0$.

n	E_0	E_1	E_2	E_3	E_4
10	$-6.3073664 \times 10^{-3}$	1.6443978×10^{-5}	$-1.6096633 \times 10^{-7}$	3.6889599×10^{-9}	$-1.6035926 \times 10^{-10}$
20	$-2.6054931 \times 10^{-3}$	1.7519735×10^{-6}	$-4.7751372 \times 10^{-9}$	$3.1028710 \times 10^{-11}$	$-3.8172037 \times 10^{-13}$
50	$-8.3108660 \times 10^{-4}$	8.9793611×10^{-8}	$-4.1940787 \times 10^{-11}$	$4.7155771 \times 10^{-14}$	$-9.9985171 \times 10^{-17}$
100	$-3.5754817 \times 10^{-4}$	9.5804953×10^{-9}	$-1.1469146 \times 10^{-12}$	$3.3159325 \times 10^{-16}$	$-1.8032908 \times 10^{-19}$

Table 3: Errors for the odd branch $r = 1$.

n	E_0	E_1	E_2	E_3	E_4
9	$-7.2226927 \times 10^{-3}$	2.3054068×10^{-5}	$-2.7222582 \times 10^{-7}$	7.4928917×10^{-9}	$-3.9120795 \times 10^{-10}$
19	$-2.7801819 \times 10^{-3}$	2.0689304×10^{-6}	$-6.2119116 \times 10^{-9}$	$4.4426296 \times 10^{-11}$	$-6.0165406 \times 10^{-13}$
49	$-8.5199643 \times 10^{-4}$	9.5862235×10^{-8}	$-4.6575971 \times 10^{-11}$	$5.4466224 \times 10^{-14}$	$-1.2012435 \times 10^{-16}$
99	$-3.6190377 \times 10^{-4}$	9.8955366×10^{-9}	$-1.2083617 \times 10^{-12}$	$3.5634509 \times 10^{-16}$	$-1.9767173 \times 10^{-19}$

14 Checks, alternative derivations, and implementation notes

14.1 Formal residual check

A generated block list should always be checked by substitution rather than by comparing displayed numerators. If

$$E_N(q) = \sum_{n=1}^{N} d_{2n-1}(\Lambda)q^{2n}, \tag{150}$$

then the exact formal certificate is

$$\Lambda E_N + \varphi(E_N) + \sum_{k=1}^{N+1} \alpha_k q^{2k}(1+E_N)^{-(2k-1)} = O(q^{2N+2}). \tag{151}$$

The coefficient of q^{2N+2} predicts the next block after division by $-\Lambda$. This identity catches sign, indexing, and centring mistakes at once.

14.2 Independent checks

The formulae in this article admit four independent comparisons.

1. Expanding the exact gamma ratio Equation (30) reproduces β_k directly.
2. Expanding the Borel kernels Equations (52) and (53) reproduces δ_k and β_k from zeta and eta values.
3. The triangular inverse recurrence and the closed Lagrange–Bell formula agree coefficient by coefficient.
4. The scaling identity in Equation (102) compares every inverse block with the independently derived half-shifted inverse-Gamma expansion in [12].

The displayed coefficients through d_{11} pass all four checks.

14.3 Truncation discipline

At recurrence step n, every formal operation may be truncated above q^{2n}. In particular, there is no reason to form a full symbolic logarithm or a full negative binomial series. A practical implementation maintains a vector of coefficients in q^2 whose entries are rational functions of Λ; multiplication is truncated convolution. The formal residual Equation (151) should be evaluated after each step.

For high n, it is often more efficient to represent a rational function in $z = \Lambda^{-1}$ as a dense polynomial and postpone common-denominator normalization. The degree bound $2n - 1$ gives an exact allocation size. The Theorem 8.3 provides a cheap additional invariant: after multiplication by $(-1)^{n+1}$, every coefficient must be positive.

15 Conclusions

The double factorial has a particularly transparent all-orders asymptotic structure once the two correct normalizations are made.

First, parity must be resolved. The branches $\mathcal{D}_0$ and $\mathcal{D}_1$ share the same gamma factor and differ only by a target constant. This isolates parity in the Lambert core and leaves a universal correction calculus. For the integer sequence, the equivalent leading amplitude is the periodic transmonomial $A_n = (2\pi)^{1/4}(\pi/2)^{(-1)^n/4}$.

Second, the shift $s = x + 1$ is the natural inversion coordinate. It converts the forward logarithm to

$$\frac{s}{2}(\log s - 1) + C_r + \text{odd inverse powers},$$

so the exact dominant inverse is $2R/W_0(2R/\mathrm{e})$ and the correction support is an even grid in the relative displacement. Every inverse block is a finite polynomial in $1/\log T$.

The coefficient problem is completely solved in two complementary ways. The triangular recurrence gives an efficient constructive algorithm, while the Lagrange–Bürmann formula in Equation (92) expresses every scalar coefficient as a finite sum of partial Bell polynomials. It yields degree and support bounds, boundary coefficients, and the coefficientwise alternating sign law. Unsigned Stirling numbers then generate the outer expansion of the Lambert core, so the complete elementary transseries has explicit combinatorial coefficients at every level.

The exact Borel kernels

$$\frac{\coth t}{2t} - \frac{1}{2t^2}, \qquad \frac{1}{2t\sinh t} - \frac{1}{2t^2}$$

complete the direct asymptotic series analytically. They give rigorous first-omitted-term bounds on the positive axis and identify the pole lattice responsible for divergence and hyperasymptotic corrections. The inverse transport formula then converts any normalized forward exponential sector into its inverse counterpart.

Finally, the single OEIS interpolation demonstrates why the interpolation choice cannot be ignored. Its periodic factor survives at leading logarithmic order and produces a bounded but rapidly oscillating $1/\log T$ sector. The closed Fourier–Lagrange formula and the periodic-grid recurrence provide its complete inverse transseries. The parity-resolved expansions are not competing answers: they are the cleaner subsequence-resummed forms appropriate when the original integer parity is known.

A Coefficient data in separated inverse-logarithmic form

For some comparisons it is useful to separate each block into powers of Λ^{-1}. The first four are

$$d_1 = \frac{1}{6\Lambda},$$

$$d_3 = -\frac{7}{180\Lambda} - \frac{1}{36\Lambda^2} - \frac{1}{72\Lambda^3}, \tag{152}$$

$$d_5 = \frac{31}{630\Lambda} + \frac{7}{270\Lambda^2} + \frac{17}{1080\Lambda^3} + \frac{5}{648\Lambda^4} + \frac{1}{432\Lambda^5}, \tag{153}$$

$$\begin{aligned} d_7 = &-\frac{127}{840\Lambda} - \frac{4063}{75600\Lambda^2} - \frac{11413}{453600\Lambda^3} - \frac{1}{80\Lambda^4} \\ &- \frac{157}{25920\Lambda^5} - \frac{35}{15552\Lambda^6} - \frac{5}{10368\Lambda^7}. \end{aligned} \tag{154}$$

The compact common-denominator forms in Equations (79) to (81) are less susceptible to transcription error; the separated form makes Equation (76) visible.

B A generic coefficient-generation pseudocode

The following pseudocode implements Equation (73) on the variable $z = q^2$.

```
input: N, symbolic Lambda
E := 0
for n := 1,...,N:
    H := (1+E)*log(1+E) - E
    for k := 1,...,n:
        H := H + alpha[k]*z^k*(1+E)^(-(2*k-1))
    H := truncate(H, z^(n+1))
    e[n] := -coefficient(H, z^n)/Lambda
    E := E + e[n]*z^n
    assert truncate(residual(E), z^(n+1)) = 0
return e[1],...,e[N]
```

All logarithms and powers here are formal truncated series. The assertion uses Equation (151).

C Notation crosswalk

Symbol	Meaning
$\mathcal{D}_r(x)$	parity-resolved gamma interpolation, $r = 0$ even and $r = 1$ odd
$\mathfrak{D}(x)$	periodic analytic continuation agreeing with all integer values
$s = x + 1$	optimal centred forward variable
C_0, C_1, C_*	even, odd, and parity-mean logarithmic constants
δ_k	uncentred logarithmic Stirling coefficients
β_k	centred logarithmic coefficients
$\alpha_k = 2\beta_k$	coefficients in the doubled inverse phase
R_r	$\log y - C_r$
w_r	$W_0(2R_r/\mathrm{e})$
T_r	exact Lambert core $2R_r/w_r$
Λ_r	core slope $w_r + 1 = \log T_r$
d_{2n-1}	branchwise inverse correction polynomial
$c_{n,j}$	coefficient of Λ^{-j} in d_{2n-1}
Δ_0, Δ_n	periodic inverse sectors for $\mathfrak{D}_+^{-1}$

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